

S/N	Question	Options	ANS
1	Formal methods are most valuable when the cost of failure is _____.	A) Low B) Moderate C) High D) Unknown	C
2	Which phase is the best place to apply model checking?	A) After deployment B) During design C) During user training D) In maintenance only	B
3	Formal verification is especially recommended for _____ algorithms.	A) Sorting B) Cryptographic C) Search D) GUI rendering	B
4	In a railway signalling system, formal methods would be used to verify _____.	A) Ticket pricing B) Train routing logic C) Station announcements D) Passenger count	B
5	Which of the following project characteristics suggests a good fit for formal methods?	A) Rapid prototyping B) Fixed, stringent safety requirements C) Ad-hoc feature requests D) Low-budget web app	B
6	Formal methods help to _____ hidden design flaws that testing may miss.	A) Hide B) Reveal C) Ignore D) Duplicate	B
7	A primary driver for adopting formal methods is _____ compliance.	A) Marketing B) Legal C) Safety D) Aesthetic	C
8	Which statement best describes the “need” for formal methods?	A) They replace all other QA activities. B) They provide mathematical assurance of correctness. C) They are only for academic use. D) They guarantee 100 % bug-free code.	B
9	In aerospace, formal methods are required to satisfy which standard?	A) ISO 27001 B) DO-178C C) ITIL D) CMMI	B
10	Formal methods reduce _____ by detecting errors early.	A) Documentation B) Post-deployment fixes C) Marketing spend D) UI design iterations	B
12	The roots of formal methods trace back to _____ in the 19th century.	A) Boolean algebra B) Object-oriented programming C) Agile Manifesto D) Cloud computing	A
12	Who introduced the first widely used formal specification language, Z, in the 1970s?	A) Edsger Dijkstra B) Sir Tony Hoare C) J. Woodcock D) Jim Woodcock & Alan Mann	D
13	The Vienna Development Method (VDM) originated in which country?	A) United States B) United Kingdom C) Austria D) Japan	C
14	Which of these tools is primarily a theorem prover?	A) SPIN B) Alloy C) Coq D) JUnit	C
15	The first practical use of model checking on a real system was on a _____ in 1981.	A) Flight control software B) Telephone switching system C) Banking transaction processor D) Word processor	B
16	Formal methods can be defined as:	A) Informal coding guidelines B) Rigorous mathematical techniques for specification & verification C) A set of UI design patterns D) A version-control system	B

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17	Which keyword best captures the essence of “formal” in this context?	A) Casual B) Approximate C) Mathematical D) Visual	C
18	Formal methods aim to prove _____ between implementation and specification.	A) Similarity B) Equivalence C) Diversity D) Redundancy	B
19	Which of the following is not a component of a formal method?	A) Formal language B) Proof system C) Code formatter D) Verification tool	C
20	The phrase “formal verification” most closely means:	A) Running many test cases B) Manually reviewing code C) Using mathematical proof to demonstrate correctness D) Checking spelling in comments	C
21	An advantage of formal methods is _____.	A) Immediate UI feedback B) Early detection of logical errors C) Faster compilation D) Reduced need for documentation	B
22	A common disadvantage is _____.	A) High upfront cost B) Unlimited scalability C) Unlimited test coverage D) No learning curve	A
23	Formal methods often _____ development time in the early phases.	A) Increase B) Decrease C) Eliminate D) Ignore	A
24	Which statement reflects a limitation?	A) They guarantee security B) They replace all testing C) They require specialized expertise D) They are only applicable to mobile apps	C
25	Which of the following is a benefit of applying formal methods to safety-critical systems?	A) Reduced certification cost B) Better aesthetic design C) Better proof of correctness D) More frequent releases	C
26	The first stage is usually _____ (creating a precise description).	A) Implementation B) Specification C) Deployment D) Maintenance	B
27	After specification, the next formal-methods stage is _____.	A) Testing B) Proof construction C) Marketing D) Documentation	B
28	Which stage involves checking a model against desired properties?	A) Validation B) Model checking C) Refactoring D) Packaging	B
29	The final stage often produces _____ that can be handed to developers.	A) Source code B) Proof obligations C) User manuals D) Wireframes	B
30	In an industrial setting, the integration of formal methods with _____ is common.	A) Agile ceremonies B) Waterfall milestones C) Continuous integration pipelines D) Social media	C